

NORTH SEA STATION 62145, United Kingdom

WMO: 994990

Lat: 53.102N

Lon: 2.800E

Elev: 30

StdP: 100.97

Time Zone: 0.00 (EUW)

Period: 02-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	0.1	1.2	-7.4	2.0	1.4	-5.7	2.3	2.2	19.4	6.8	18.3	6.6	7.7	90	0.753

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.3	21.4	17.3	20.3	16.9	19.5	16.5	18.2	20.2	17.6	19.6	17.0	19.1	5.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.5	12.6	19.2	16.7	12.0	18.6	16.0	11.4	18.4	51.8	20.3	49.5	19.6	47.8	19.1	22.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.3	15.2	13.8	0.1	25.3	1.4	2.1	-0.9	26.8	-1.8	28.0	-2.6	29.1	-3.6	30.6		
(5)			-2.1	18.9	1.9	0.8	-3.5	19.4	-4.6	19.9	-5.7	20.3	-7.1	20.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.1	6.1	5.1	6.2	8.6	11.3	13.9	16.5	17.5	16.1	13.3	10.6	7.5
	DBStd	4.72	2.13	2.32	2.38	2.27	2.20	2.13	1.89	1.87	2.03	2.29	2.07	2.36
	HDD10.0	542	120	137	118	54	12	1	0	0	0	3	17	82
	HDD18.3	2664	378	370	375	292	219	133	63	39	71	156	231	336
	CDD10.0	944	0	0	1	12	52	118	201	232	184	106	36	4
	CDD18.3	24	0	0	0	0	0	1	6	12	6	1	0	0
	CDH23.3	9	0	0	0	0	0	1	5	3	1	0	0	0
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0
Wind	WSAvg	6.9	9.0	7.5	7.1	5.8	6.2	5.5	5.7	5.9	6.3	7.3	8.2	8.7
Precipitation	PrecAvg	975	94	72	66	46	52	63	74	93	95	104	108	108
	PrecMax	1262	165	146	139	99	88	135	172	209	192	191	192	182
	PrecMin	724	12	23	10	3	24	19	13	20	19	37	18	50
	PrecStd	132	39	31	31	24	19	30	36	46	43	43	39	35
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.2	10.6	12.1	15.2	17.8	20.7	23.2	23.6	22.1	19.7	16.7	12.7
		MCWB	9.4	8.3	8.9	11.1	13.4	16.4	17.8	18.5	17.9	16.3	14.2	10.8
	2%	DB	10.4	9.7	10.4	13.3	16.1	18.8	20.8	21.5	20.4	18.0	15.1	12.1
		MCWB	8.9	7.8	8.0	10.2	12.7	15.2	16.8	17.5	17.1	15.8	12.7	10.2
	5%	DB	9.7	9.0	9.6	12.2	14.8	17.3	19.6	20.6	19.4	17.2	14.2	11.5
		MCWB	8.2	7.3	7.5	9.6	12.0	14.6	16.5	17.2	16.7	15.0	12.0	9.6
	10%	DB	9.0	8.2	8.9	11.2	13.7	16.2	18.6	19.7	18.6	16.2	13.5	10.8
		MCWB	7.4	6.6	7.0	9.0	11.4	14.1	16.0	16.8	16.2	13.9	11.4	8.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.9	9.1	9.4	11.7	14.7	17.5	18.9	19.3	18.6	17.2	14.2	11.5
		MCDB	10.8	10.0	11.4	14.6	16.4	19.2	21.9	21.2	20.2	18.8	16.0	12.5
	2%	WB	9.1	8.2	8.4	10.6	13.2	16.1	17.7	18.5	17.8	16.0	13.2	10.5
		MCDB	10.1	9.4	9.9	12.8	15.2	17.7	19.9	20.5	19.6	17.7	14.8	11.8
	5%	WB	8.4	7.5	7.8	9.9	12.3	15.1	16.9	17.8	17.1	15.3	12.4	9.7
		MCDB	9.5	8.7	9.2	11.8	14.2	16.7	19.2	19.8	19.1	16.9	13.9	11.2
	10%	WB	7.6	6.8	7.3	9.2	11.7	14.4	16.1	17.1	16.5	14.4	11.7	9.1
		MCDB	8.7	8.1	8.7	11.0	13.5	15.9	18.2	19.2	18.6	16.1	13.1	10.5
Mean Daily Temperature Range	5% DB	MDBR	2.3	2.2	2.7	3.3	3.4	3.4	3.5	3.3	2.9	2.5	2.5	2.4
		MCDBR	2.9	2.6	3.6	4.9	5.4	5.5	5.4	5.1	4.4	3.4	3.1	2.7
		MCWBR	3.4	2.5	2.6	2.9	3.3	3.5	2.9	3.0	3.1	3.3	3.2	3.2
	5% WB	MCDBR	2.8	2.6	3.3	4.5	4.8	5.2	4.8	4.4	4.2	3.1	2.9	2.7
		MCWBR	3.4	2.5	2.6	2.8	3.2	3.6	2.7	3.0	3.2	3.5	3.3	3.3
Clear-Sky Solar Irradiance	taub	0.332	0.348	0.378	0.403	0.402	0.405	0.409	0.398	0.384	0.362	0.337	0.321	
	taud	2.333	2.369	2.313	2.281	2.359	2.361	2.360	2.401	2.431	2.429	2.375	2.322	
	Ebn at Noon	636	751	800	828	849	848	836	825	790	725	625	572	
	Edh at Noon	61	79	101	118	115	116	114	104	90	74	58	52	
All-Sky Solar Radiation	RadAvg	0.65	1.37	2.70	4.47	5.54	5.81	5.60	4.71	3.27	1.75	0.79	0.49	
	RadStd	0.06	0.23	0.24	0.42	0.35	0.31	0.58	0.33	0.24	0.14	0.07	0.05	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page